

Tiktok_Project

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1 TikTok Project - César Rábago Pérez

Course 2 - Get Started with Python

Welcome to the TikTok Project!

You have just started as a data professional at TikTok.

The team is still in the early stages of the project. You have received notice that TikTok's leadership team has approved the project proposal. To gain clear insights to prepare for a claims classification model, TikTok's provided data must be examined to begin the process of exploratory data analysis (EDA).

A notebook was structured and prepared to help you in this project. Please complete the following questions.

2 Course 2 End-of-course project: Inspect and analyze data

In this activity, you will examine data provided and prepare it for analysis.

The purpose of this project is to investigate and understand the data provided. This activity will:

1. Acquaint you with the data
2. Compile summary information about the data
3. Begin the process of EDA and reveal insights contained in the data
4. Prepare you for more in-depth EDA, hypothesis testing, and statistical analysis

The goal is to construct a dataframe in Python, perform a cursory inspection of the provided dataset, and inform TikTok data team members of your findings. *This activity has three parts:*

Part 1: Understand the situation * How can you best prepare to understand and organize the provided TikTok information?

Part 2: Understand the data

- Create a pandas dataframe for data learning and future exploratory data analysis (EDA) and statistical activities
- Compile summary information about the data to inform next steps

Part 3: Understand the variables

- Use insights from your examination of the summary data to guide deeper investigation into variables

To complete the activity, follow the instructions and answer the questions below. Then, you will use your responses to these questions and the questions included in the Course 2 PACE Strategy Document to create an executive summary.

Be sure to complete this activity before moving on to Course 3. You can assess your work by comparing the results to a completed exemplar after completing the end-of-course project.

3 Identify data types and compile summary information

Throughout these project notebooks, you'll see references to the problem-solving framework PACE. The following notebook components are labeled with the respective PACE stage: Plan, Analyze, Construct, and Execute.

4 PACE stages

- [Plan] (#scrollTo=psz51YkZVwtN&line=3&uniquifier=1)
- [Analyze] (#scrollTo=mA7Mz_SnI8km&line=4&uniquifier=1)
- [Construct] (#scrollTo=Lca9c8XON8lc&line=2&uniquifier=1)
- [Execute] (#scrollTo=401PgchTPr4E&line=2&uniquifier=1)

4.1 PACE: Plan

Consider the questions in your PACE Strategy Document and those below to craft your response:

4.1.1 Task 1. Understand the situation

- How can you best prepare to understand and organize the provided information?

Review the data dictionary to understand what each variable means.

Load the dataset into a dataframe for easy inspection and analysis.

Check data quality by identifying missing values, data types, and inconsistencies.

Explore summary statistics and sample rows to get a sense of typical values versus outliers.

Organize variables into categories (e.g., content attributes, author attributes, engagement metrics) to structure the analysis.

4.2 PACE: Analyze

Consider the questions in your PACE Strategy Document to reflect on the Analyze stage.

4.2.1 Task 2a. Imports and data loading

Start by importing the packages that you will need to load and explore the dataset. Make sure to use the following import statements: `* import pandas as pd`

- `import numpy as np`

```
[1]: import pandas as pd

import numpy as np
```

Then, load the dataset into a dataframe. Creating a dataframe will help you conduct data manipulation, exploratory data analysis (EDA), and statistical activities.

Note: As shown in this cell, the dataset has been automatically loaded in for you. You do not need to download the .csv file, or provide more code, in order to access the dataset and proceed with this lab. Please continue with this activity by completing the following instructions.

```
[2]: # Load dataset into dataframe
data = pd.read_csv("tiktok_dataset.csv")
```

4.2.2 Task 2b. Understand the data - Inspect the data

View and inspect summary information about the dataframe by **coding the following:**

1. `data.head(10)`
2. `data.info()`
3. `data.describe()`

Consider the following questions:

Question 1: When reviewing the first few rows of the dataframe, what do you observe about the data? What does each row represent?

Question 2: When reviewing the `data.info()` output, what do you notice about the different variables? Are there any null values? Are all of the variables numeric? Does anything else stand out?

Question 3: When reviewing the `data.describe()` output, what do you notice about the distributions of each variable? Are there any questionable values? Does it seem that there are outlier values?

```
[3]: # Display and examine the first ten rows of the dataframe
data.head(10)
```

```
[3]:      # claim_status      video_id  video_duration_sec  \
0      1          claim  7017666017          59
1      2          claim  4014381136          32
2      3          claim  9859838091          31
3      4          claim  1866847991          25
4      5          claim  7105231098          19
5      6          claim  8972200955          35
6      7          claim  4958886992          16
7      8          claim  2270982263          41
8      9          claim  5235769692          50
9     10          claim  4660861094          45
```

```
      video_transcription_text verified_status  \
0  someone shared with me that drone deliveries a...  not verified
1  someone shared with me that there are more mic...  not verified
2  someone shared with me that american industria...  not verified
3  someone shared with me that the metro of st. p...  not verified
4  someone shared with me that the number of busi...  not verified
5  someone shared with me that gross domestic pro...  not verified
6  someone shared with me that elvis presley has ...  not verified
7  someone shared with me that the best selling s...  not verified
8  someone shared with me that about half of the ...  not verified
9  someone shared with me that it would take a 50...  verified
```

```
      author_ban_status  video_view_count  video_like_count  video_share_count  \
0      under review      343296.0      19425.0      241.0
1      active      140877.0      77355.0      19034.0
2      active      902185.0      97690.0      2858.0
3      active      437506.0      239954.0      34812.0
4      active      56167.0      34987.0      4110.0
5      under review      336647.0      175546.0      62303.0
6      active      750345.0      486192.0      193911.0
7      active      547532.0      1072.0      50.0
8      active      24819.0      10160.0      1050.0
9      active      931587.0      171051.0      67739.0
```

```
      video_download_count  video_comment_count
0          1.0          0.0
1        1161.0         684.0
2         833.0         329.0
3        1234.0         584.0
4         547.0         152.0
5        4293.0        1857.0
6        8616.0        5446.0
7          22.0          11.0
8          53.0          27.0
9       4104.0       2540.0
```

```
[4]: # Get summary info
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19382 entries, 0 to 19381
Data columns (total 12 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   #                                     19382 non-null  int64
1   claim_status                         19084 non-null  object
2   video_id                             19382 non-null  int64
3   video_duration_sec                   19382 non-null  int64
4   video_transcription_text             19084 non-null  object
5   verified_status                      19382 non-null  object
6   author_ban_status                   19382 non-null  object
7   video_view_count                     19084 non-null  float64
8   video_like_count                     19084 non-null  float64
9   video_share_count                    19084 non-null  float64
10  video_download_count                 19084 non-null  float64
11  video_comment_count                  19084 non-null  float64
dtypes: float64(5), int64(3), object(4)
memory usage: 1.8+ MB
```

```
[5]: # Get summary statistics
data.describe()
```

```
[5]:
```

	#	video_id	video_duration_sec	video_view_count	\
count	19382.000000	1.938200e+04	19382.000000	19084.000000	
mean	9691.500000	5.627454e+09	32.421732	254708.558688	
std	5595.245794	2.536440e+09	16.229967	322893.280814	
min	1.000000	1.234959e+09	5.000000	20.000000	
25%	4846.250000	3.430417e+09	18.000000	4942.500000	
50%	9691.500000	5.618664e+09	32.000000	9954.500000	
75%	14536.750000	7.843960e+09	47.000000	504327.000000	
max	19382.000000	9.999873e+09	60.000000	999817.000000	

	video_like_count	video_share_count	video_download_count	\
count	19084.000000	19084.000000	19084.000000	
mean	84304.636030	16735.248323	1049.429627	
std	133420.546814	32036.174350	2004.299894	
min	0.000000	0.000000	0.000000	
25%	810.750000	115.000000	7.000000	
50%	3403.500000	717.000000	46.000000	
75%	125020.000000	18222.000000	1156.250000	
max	657830.000000	256130.000000	14994.000000	


```
video_comment_count
```

count	19084.000000
mean	349.312146
std	799.638865
min	0.000000
25%	1.000000
50%	9.000000
75%	292.000000
max	9599.000000

Question 1: Each row represents a TikTok video with content attributes (claim/opinion, duration, transcription) and engagement metrics (views, likes, shares, comments).

Question 2: There are missing values in some columns (claim_status, transcription, and engagement metrics). The dataset includes both numeric and categorical variables, and not all records are complete.

Question 3: The metrics show skewed distributions with outliers: most videos have low views/likes, but a few reach extremely high values.

4.2.3 Task 2c. Understand the data - Investigate the variables

In this phase, you will begin to investigate the variables more closely to better understand them.

You know from the project proposal that the ultimate objective is to use machine learning to classify videos as either claims or opinions. A good first step towards understanding the data might therefore be examining the `claim_status` variable. Begin by determining how many videos there are for each different claim status.

```
[8]: # What are the different values for claim status and how many of each are in
      ↪ the data?

claim_counts = data['claim_status'].value_counts(dropna=False)
print("Number of videos per claim status:\n", claim_counts)
```

Number of videos per claim status:

claim	9608
opinion	9476
NaN	298

Name: claim_status, dtype: int64

Question: What do you notice about the values shown?

Next, examine the engagement trends associated with each different claim status.

Start by using Boolean masking to filter the data according to claim status, then calculate the mean and median view counts for each claim status.

```
[9]: # What is the average view count of videos with "claim" status?
      # Filter the dataset for videos with claim_status == "claim"
claim_videos = data[data['claim_status'] == 'claim']
```

```
# Calculate the average view count
average_views_claim = claim_videos['video_view_count'].mean()
print("Average view count for 'claim' videos:", average_views_claim)
```

Average view count for 'claim' videos: 501029.4527477102

```
[10]: # Filter the dataset for videos with claim_status == "opinion"
opinion_videos = data[data['claim_status'] == 'opinion']

# Calculate the average view count
average_views_opinion = opinion_videos['video_view_count'].mean()
print("Average view count for 'opinion' videos:", average_views_opinion)
```

Average view count for 'opinion' videos: 4956.43224989447

Question: What do you notice about the mean and media within each claim category?

Now, examine trends associated with the ban status of the author.

Use `groupby()` to calculate how many videos there are for each combination of categories of claim status and author ban status.

```
[1]: import pandas as pd
import numpy as np

# Load dataset into dataframe
data = pd.read_csv("tiktok_dataset.csv")

# Get counts for each group combination of claim status and author ban status
ban_status_counts = data.groupby(["claim_status", "author_ban_status"]).size()

print("Number of videos for combination of claim_status y author_ban_status:\n")
print(ban_status_counts)
```

Number of videos for combination of claim_status y author_ban_status:

claim_status	author_ban_status	
claim	active	6566
	banned	1439
	under review	1603
opinion	active	8817
	banned	196
	under review	463

dtype: int64

Question: What do you notice about the number of claims videos with banned authors? Why might this relationship occur?

Continue investigating engagement levels, now focusing on `author_ban_status`.

Calculate the median video share count of each author ban status.

```
[8]: # Median video share count per author ban status
median_shares = data.groupby("author_ban_status")["video_share_count"].median()

print("Median video share count by author ban status:\n")
print(median_shares)
```

Median video share count by author ban status:

```
author_ban_status
active          437.0
banned         14468.0
under review    9444.0
Name: video_share_count, dtype: float64
```

```
[9]: # What's the median video share count of each author ban status?

median_shares = data.groupby("author_ban_status")["video_share_count"].median()

print("Median video share count by author ban status:\n")
print(median_shares)
```

Median video share count by author ban status:

```
author_ban_status
active          437.0
banned         14468.0
under review    9444.0
Name: video_share_count, dtype: float64
```

Question: What do you notice about the share count of banned authors, compared to that of active authors? Explore this in more depth.

Use `groupby()` to group the data by `author_ban_status`, then use `agg()` to get the count, mean, and median of each of the following columns: * `video_view_count` * `video_like_count` * `video_share_count`

Remember, the argument for the `agg()` function is a dictionary whose keys are columns. The values for each column are a list of the calculations you want to perform.

```
[10]: # Group by author_ban_status and calculate count, mean, and median for selected
      ↪ columns
stats_by_ban_status = data.groupby("author_ban_status").agg({
    "video_view_count": ["count", "mean", "median"],
    "video_like_count": ["count", "mean", "median"],
    "video_share_count": ["count", "mean", "median"]
})
```



```
# Display the results
print("Video statistics by author ban status:\n")
print(stats_by_ban_status)
```

Video statistics by author ban status:

	video_view_count			video_like_count	
	count	mean	median	count	
author_ban_status					
active	15383	215927.039524	8616.0	15383	
banned	1635	445845.439144	448201.0	1635	
under review	2066	392204.836399	365245.5	2066	

		video_share_count		
	mean	median	count	mean
author_ban_status				
active	71036.533836	2222.0	15383	14111.466164
banned	153017.236697	105573.0	1635	29998.942508
under review	128718.050339	71204.5	2066	25774.696999

	median
author_ban_status	
active	437.0
banned	14468.0
under review	9444.0

Question: What do you notice about the number of views, likes, and shares for banned authors compared to active authors?

Now, create three new columns to help better understand engagement rates: * `likes_per_view`: represents the number of likes divided by the number of views for each video * `comments_per_view`: represents the number of comments divided by the number of views for each video * `shares_per_view`: represents the number of shares divided by the number of views for each video

```
[13]: # Create new engagement rate columns
data["likes_per_view"] = data["video_like_count"] / data["video_view_count"]
data["comments_per_view"] = data["video_comment_count"] /
    ↳data["video_view_count"]
data["shares_per_view"] = data["video_share_count"] / data["video_view_count"]

# Preview the first few rows to check the new columns
data[["video_like_count", "video_comment_count", "video_share_count",
      "video_view_count", "likes_per_view", "comments_per_view",
      ↳"shares_per_view"]].head()
```

```
[13]: video_like_count  video_comment_count  video_share_count  video_view_count  \
0          19425.0             0.0          241.0          343296.0
```

1	77355.0	684.0	19034.0	140877.0
2	97690.0	329.0	2858.0	902185.0
3	239954.0	584.0	34812.0	437506.0
4	34987.0	152.0	4110.0	56167.0

	likes_per_view	comments_per_view	shares_per_view
0	0.056584	0.000000	0.000702
1	0.549096	0.004855	0.135111
2	0.108282	0.000365	0.003168
3	0.548459	0.001335	0.079569
4	0.622910	0.002706	0.073175

Use `groupby()` to compile the information in each of the three newly created columns for each combination of categories of claim status and author ban status, then use `agg()` to calculate the count, the mean, and the median of each group.

```
[16]: # Group by claim_status and author_ban_status, then calculate stats for
      ↪ engagement rates
engagement_stats = data.groupby(["claim_status", "author_ban_status"]).agg({
    "likes_per_view": ["count", "mean", "median"]
})

# Display the result
print("Engagement rates by claim status and author ban status:\n")
print(engagement_stats)
```

Engagement rates by claim status and author ban status:

		likes_per_view		
		count	mean	median
claim_status	author_ban_status			
claim	active	6566	0.329542	0.326538
	banned	1439	0.345071	0.358909
	under review	1603	0.327997	0.320867
opinion	active	8817	0.219744	0.218330
	banned	196	0.206868	0.198483
	under review	463	0.226394	0.228051

Question:

How does the data for claim videos and opinion videos compare or differ? Consider views, comments, likes, and shares.

A= Claim content is both more viral and more engaging per view, making it a stronger driver of platform activity, but potentially also a moderation risk.

4.3 PACE: Construct

Note: The Construct stage does not apply to this workflow. The PACE framework can be adapted to fit the specific requirements of any project.

4.4 PACE: Execute

Consider the questions in your PACE Strategy Document and those below to craft your response.

4.4.1 Given your efforts, what can you summarize for Rosie Mae Bradshaw and the TikTok data team?

Note for Learners: Your answer should address TikTok's request for a summary that covers the following points:

- What percentage of the data is comprised of claims and what percentage is comprised of opinions?
- What factors correlate with a video's claim status?
- What factors correlate with a video's engagement level?
- Claim videos: 9,608 out of 19,382 total → ~49.6%

Opinion videos: 9,476 out of 19,382 total → ~48.9%

Unspecified (NaN): 298 out of 19,382 total → ~1.5%

- A video's claim status is most strongly associated with higher visibility (views), stronger engagement (likes and shares per view), and increased likelihood of being tied to banned or reviewed authors.
- Engagement levels are most strongly tied to:

Whether a video is a claim (claims attract far more engagement than opinions).

The author's ban status (banned/under review accounts see disproportionately high engagement).

Engagement efficiency ratios (likes-per-view, shares-per-view).